GenoFold: A Multi-Modal Graph–Transformer Reconstructor of 3D Genome Architecture with Per-Bin Uncertainty

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***Abstract*—Most methods that rebuild chromosome structure from Hi-C use the contact matrix alone and solve each matrix from scratch. GenoFold is a trained reconstructor whose genomic bins carry sequencing coverage, 4-mer composition, GC content and mappability alongside Hi-C contacts; a four-layer GATv2 encoder and a six-layer Transformer feed a coordinate decoder trained with a self-supervised distance-matching loss. On dense GM12878 Hi-C it does not beat existing methods: its distance Spearman correlation (dSCC) on chromosome 22 is 0.893 ± 0.001, against 0.900 for 3DMax and 0.889 for HiC-GNN, and the gaps are of the order of run-to-run variation. Scored against distances measured by chromatin tracing in IMR90 and K562, GenoFold reaches Spearman 0.986 and 0.981. When contacts are downsampled to 5%, it still reaches 0.978 and 0.969 while 3DMax falls to 0.819 and 0.831. A modality ablation shows that only a small part of this gap comes from the extra node features; most of it appears to come from the structural prior the trained decoder has learned. A per-bin uncertainty estimate ranks reconstruction error only weakly (Spearman 0.28), and we report it as such.**

***Index Terms*—3D genome reconstruction, Hi-C, graph attention networks, multi-modal learning, uncertainty estimation**

I. Introduction

Chromatin folding decides which enhancers can contact their promoters and where topologically associating domains (TADs) begin and end [1], [2]. Hi-C measures how often pairs of loci are ligated together [3], and a family of methods converts those frequencies into coordinates. 3DMax optimizes coordinates by maximum likelihood with contact frequency standing in for distance [4]; ShRec3D embeds shortest-path distances by multidimensional scaling [5]; FLAMINGO completes sparse high-resolution matrices with a low-rank model [6]; HiC-GNN learns node embeddings with a graph neural network [7]. All four read Hi-C and nothing else, and all but HiC-GNN fit each matrix independently, so a shallow library gives them little to work with.

GenoFold [8] tests whether two additions help: node features that do not depend on sequencing depth (coverage, sequence composition, GC content and mappability), and a trained network that can carry structure learned on one matrix into another. Our claims are narrower than in the submitted version. We contribute (i) a trained GATv2–Transformer reconstructor that matches established methods on dense Hi-C; (ii) validation against measured chromatin-tracing distances, which are independent of the training signal; (iii) a sparse-contact experiment in which GenoFold degrades far less than per-matrix optimizers, together with an ablation that attributes most of that robustness to the learned prior rather than to the added modalities; and (iv) a per-bin uncertainty estimate and feature-group attributions, whose limits we state.

TABLE I
Node features ($F = 140$). Only the coverage track depends on sequencing depth; contacts enter through the edges.

Group	Dim	Source
Coverage	1	RPKM read depth, GC/mappability-corrected
Sequence	136	RC-collapsed 4-mer frequencies (hg19)
Metadata	2	GC content; CRG 50-mer mappability
Position	1	Normalized bin index

II. Method

Preprocessing. Bins whose row-wise zero fraction exceeds 0.97 are removed; the threshold is looser than the usual 0.8 so that sparse pericentromeric bins are kept. Matrices are balanced by ICE (tolerance 10^{-5}) [9]. Coverage is RPKM-normalized, corrected for GC bias and mappability, and z-scored; GC content comes from hg19 and mappability from the ENCODE/CRG 50-mer track.

Graph. Each chromosome becomes a graph with M nodes. A node carries 140 features: coverage (1), reverse-complement-collapsed 4-mer frequencies (136), GC and mappability (2), and normalized position (1). Edges carry the log contact weight and log genomic distance, stored as a PyTorch Geometric object [10].

Encoder and decoder. Four GATv2 layers [11] (8 heads with residual connections in the first three, 4 averaged heads in the last, $d = 64$) supply local context. Six pre-norm Transformer layers [12] (8 heads, FFN width 512, GELU, dropout 0.1) add interactions between all pairs of bins. A five-layer MLP ($64 \to 128 \to 256 \to 128 \to 3$) decodes coordinates. The code

also contains a species embedding; every experiment here uses one species, so the embedding holds a single constant value and has no effect. We do not count it as part of the method. The model has 2.71 M parameters.

Objective. Contacts map to wish distances $D_{ij} = C_{ij}^{-0.25}$, and the network minimizes

$$\mathcal{L} = \frac{1}{|P|} \sum_{(i,j)\in P} \left(\hat{z}_{ij} - z^*_{ij}\right)^2 + \lambda\, \mathcal{L}_{\mathrm{NLL}}, \tag{1}$$

where $\hat{z}_{ij}$ and z^*_{ij} are z-scored log reconstructed and wish distances over observed pairs P, and $\mathcal{L}_{\mathrm{NLL}}$ is a Gaussian negative log-likelihood that trains the variance head. We use Adam (learning rate 10^{-3}, cosine schedule, weight decay 10^{-5}) for 300 epochs. Held-out dSCC on a 20% pair holdout levels off by epoch 150 and moves by less than 0.002 over the last 50 epochs. No early stopping is used; all results come from the epoch-300 checkpoint. Held-out dSCC (0.890) and training dSCC (0.890) differ by less than 0.001, so the model is not fitting individual pairs. A per-matrix optimizer such as 3DMax has no counterpart to this test, since it fits every observed pair by construction.

Uncertainty. The variance head predicts per-axis log-variances v_{ik}; T Monte Carlo dropout passes [13] add an epistemic term. The per-bin radius is $r_i = (\sum_k \sigma_{ik}^2)^{1/2}$ with $\sigma_{ik} = \exp(v_{ik}/2)$.

Attribution. Integrated Gradients [14] (32 steps from a zero baseline) are summed within each of the four feature groups. TADs are called on reconstructions by insulation score [15].

III. Experimental Setup

We use the Rao et al. *in situ* Hi-C maps for GM12878, IMR90 and K562 (GSE63525, KR-normalized) [3]. The main benchmark is GM12878 chromosome 22 at 50 kb (701 bins after filtering), scored by dSCC, the Spearman correlation between reconstructed and wish distances over observed pairs, and by TAD-boundary F_1. 3DMax, HiC-GNN and ShRec3D are re-implemented on the identical submatrix; FLAMINGO runs from its published R package. Means and standard deviations are over three seeds.

For a check that does not reuse the training signal we take the chromatin-tracing data of Bintu et al. [16]: median pairwise distances between 65 consecutive 30 kb segments of chromosome 21 (hg19 chr21:29.37–31.32 Mb), imaged in thousands of IMR90 and K562 cells. Each method reconstructs the region from the matched 10 kb Hi-C; coordinates are coarse-grained onto the imaging grid and correlated with the imaged medians. The tracing coordinates are annotated on hg38; we register them to the hg19 Hi-C by the liftOver offset (+1.37 Mb) that maximizes agreement between raw Hi-C and imaging, and apply the same registration to every method. For the sparse experiments, each contact matrix is thinned by keeping a random fraction of its nonzero bin pairs (values unchanged, symmetry preserved); each method then reconstructs from the thinned matrix alone. This removes pairs rather than reads, so it mimics loss of coverage more crudely than read-level subsampling.

TABLE II
Native-resolution comparison at 5 kb on chr22:20–30 Mb (1,964 bins). Short-range: pairs ≤1 Mb apart. *FLAMINGO from its R package, 3 seeds; others single runs.

Method	Global dSCC	Short-range dSCC
3DMax	**0.820**	**0.895**
GenoFold	0.814	0.892
ShRec3D	0.783	0.677
FLAMINGO*	0.655 ± 0.017	0.667

TABLE III
GM12878 chr22, 50 kb, 701 bins (mean ± s.d., 3 seeds; ShRec3D is deterministic). Time is CPU wall-clock per run.

Method	dSCC	TAD F_1	Time (s)
ShRec3D	0.887	0.45	1.4
3DMax	$\mathbf{0.900} \pm 0.000$	0.59 ± 0.01	29.1
HiC-GNN	0.889 ± 0.002	$\mathbf{0.60} \pm 0.01$	100.6
GenoFold	0.893 ± 0.001	0.57 ± 0.01	244.9

IV. Results

A. Dense Hi-C

Table III gives the GM12878 benchmark. 3DMax has the highest dSCC (0.900); GenoFold (0.893) and HiC-GNN (0.889) follow, and the differences are of the same size as the 0.01–0.02 spread we see between runs. We do not read them as a ranking. GenoFold is also the slowest method (245 s on CPU), most of it spent in the $O(M^2)$ Transformer attention and the dropout passes.

FLAMINGO is built for 1–5 kb data, and at 50 kb its dSCC is 0.27, which says more about the setting than the method; we therefore leave it out of Table III. At its native 5 kb, on chr22:20–30 Mb (1,964 bins), FLAMINGO reaches 0.655 (Table II), against 0.820 for 3DMax, 0.814 for GenoFold and 0.783 for ShRec3D. These dSCC numbers favour the all-pair optimizers, since dSCC is close to their objective and not to FLAMINGO's.

Other chromosomes. The full baseline suite has so far been run only on chromosome 22. For GenoFold alone, whole-chromosome runs give dSCC 0.891 on chr22 (701 bins), 0.880 on chr20 (1,196) and 0.865 on chr16 (1,582); runtime rises from 223 s to 1,218 s and peak memory from 1.7 to 3.7 GB (Table IV). Accuracy therefore declines slowly as chromosomes grow, but without matching baseline runs these numbers say nothing about relative standing. A per-chromosome comparison across all methods is the first item of our follow-up work.

Fig. 1 shows the chr22 reconstruction next to the observed contact map. The pseudo-contacts implied by the coordinates reproduce the block structure of the observed map, though the reconstruction smooths the sharpest domain corners.

B. Agreement with Measured Distances

Table V compares reconstructed and imaged distances. At full depth GenoFold (0.986, 0.981) and 3DMax (0.987, 0.979) are level, and both exceed the raw Hi-C contacts read directly

TABLE IV
GENOFOLD ON WHOLE CHROMOSOMES OF GM12878 AT 50 KB (CPU).

Chrom.	Bins	dSCC	Time (s)	Peak RSS (GB)
chr22	701	0.891	223	1.7
chr20	1,196	0.880	632	2.8
chr16	1,582	0.865	1,218	3.7

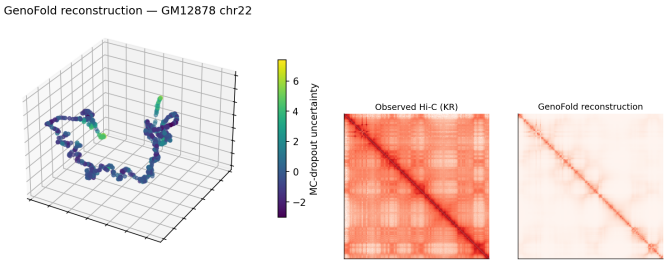


Fig. 1. GM12878 chr22. Left: reconstruction coloured by per-bin uncertainty. Right: observed KR-normalized contacts against the pseudo-contacts implied by the reconstruction ($\log(1+x)$).

as distances (0.973, 0.963). Because the imaging data were never used for training, this agreement is evidence that the dSCC scores reflect physical structure and not only consistency with the loss.

C. Sparse Hi-C, and What Drives It

We downsampled each cell line's contacts to 25%, 10% and 5%, reconstructed from the thinned matrix, and scored against the full-depth imaging data (Fig. 2, Table V). GenoFold changes by less than 0.02 across this range. 3DMax holds up to 10% (0.955, 0.948) and then drops to 0.819 and 0.831 at 5%; ShRec3D drops further. Below 10% GenoFold's reconstructions agree with the imaging data better than the thinned contacts themselves do.

Both reviewers asked whether this gap comes from the added modalities or simply from having a trained model. Our own ablation points to the second. On GM12878 chr22, a GenoFold variant with Hi-C-only nodes trails the full model by only 0.002 dSCC at 50% depth and 0.005 at 5% (0.884 against 0.879), and at full depth every configuration lies within 0.01 (Hi-C only 0.889; without 4-mers 0.886; without GC and mappability 0.894). Table VI lists these runs. A margin that small cannot account for a 0.14–0.16 lead over 3DMax. The more likely explanation is that the decoder has learned what chromatin at 10–50 kb tends to look like, which a per-matrix optimizer cannot do. Two experiments would settle the question, and we have not yet run either: the imaging comparison with a Hi-C-only GenoFold, and the same comparison with HiC-GNN, the one trainable baseline, which is missing from the sparse experiment. Until they are done we state the sparse result as an advantage of a trained reconstructor, not of multimodality.

D. Transfer Across Cell Lines

Because GenoFold learns a mapping rather than a set of free coordinates, a model trained on one matrix can be applied to another. Retrained on IMR90 and K562 chr22, it reaches dSCC 0.795 and 0.754 (Table VII). These scores are lower than on GM12878 mainly because the IMR90 and K562 libraries have roughly a fifth of GM12878's read depth at this locus. A GM12878-trained model applied without retraining gives 0.630 and 0.653: well above chance, but clearly below retraining. Transfer is therefore a convenience for a quick first structure, and retraining remains the better choice when a target matrix is available.

TABLE V
SPEARMAN CORRELATION WITH MEDIAN CHROMATIN-TRACING DISTANCES (CHR21, 65 SEGMENTS) [16] AT FULL HI-C DEPTH AND WITH 5% OF CONTACTS KEPT. GENOFOLD: MEAN OF 3 SEEDS (S.D. $\leq$0.004). RAW HI-C SCORES THE (THINNED) CONTACTS AS DISTANCES, WITH NO RECONSTRUCTION.

	IMR90		K562	
Method	100%	5%	100%	5%
GenoFold	0.986	**0.978**	**0.981**	**0.969**
3DMax	**0.987**	0.819	0.979	0.831
ShRec3D	0.970	0.689	0.960	0.581
Raw Hi-C	0.973	0.904	0.963	0.885

TABLE VI
MODALITY ABLATION, GM12878 CHR22 (HELD-OUT DSCC). LEFT: FULL DEPTH; "SPARSE" RESTRICTS TO PAIRS TOUCHING THE LOWEST-COVERAGE THIRD OF BINS. RIGHT: HI-C-ONLY VERSUS FULL MODEL AS CONTACTS ARE REMOVED.

Configuration	All pairs	Sparse
Full model	0.890	0.871
− 4-mers	0.886	0.867
− GC, mappability	0.894	0.877
Hi-C only	0.889	0.871

Depth	Full	Hi-C only
50%	0.890	0.888
20%	0.889	0.887
10%	0.892	0.888
5%	0.884	0.879

E. Uncertainty

The combined per-bin radius correlates with per-bin reconstruction error at Spearman 0.28 on chr22. Fig. 3 shows how that correlation arises. Sorting bins into deciles of predicted uncertainty, the mean error stays between 0.087 and 0.099 over the lowest nine deciles and rises to 0.126 in the highest. The estimate thus flags the worst tenth of bins and does little ranking among the rest. We have not compared it with a simple proxy such as per-bin contact count, nor tested whether the radii give the coverage their nominal intervals imply. Until those tests exist the radius should be read as a coarse warning signal; we no longer describe it as calibrated.

F. Attribution

Integrated Gradients on the trained chr22 model, summed by feature group and normalized (Fig. 4), give 0.43 to GC and mappability, 0.30 to coverage, 0.18 to position and 0.10 to the 136 4-mer features. Two cautions apply. Attribution measures sensitivity, and the ablation shows that removing GC and mappability does not lower dSCC (0.894 against 0.890). The two results are compatible: the model reads these features, but at full depth the contact graph carries the same information.

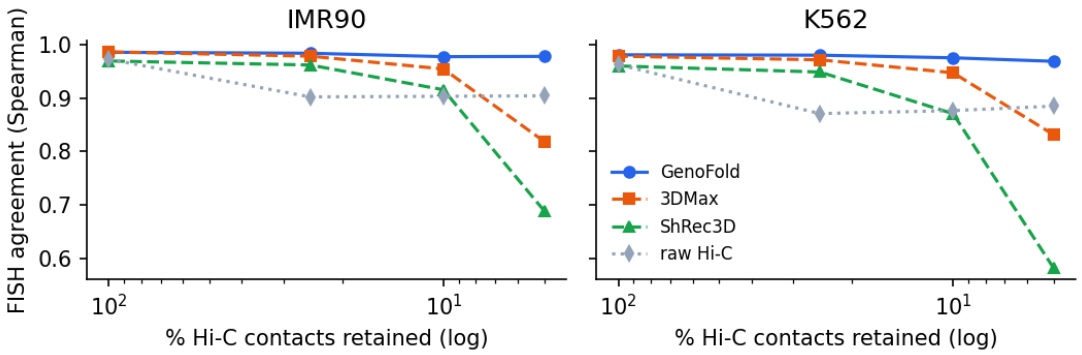


Fig. 2. Agreement with imaged chr21 distances as Hi-C contacts are downsampled (log axis, right to left) in IMR90 and K562. GenoFold stays within 0.02 of its full-depth score; 3DMax and ShRec3D fall sharply below 10% of contacts. Raw Hi-C scores the thinned contacts as distances.

TABLE VII
CROSS-CELL-LINE RESULTS ON CHR22. TRANSFER APPLIES THE GM12878-TRAINED MODEL UNCHANGED; RETRAINED FITS THE TARGET LINE.

Cell line	Transfer dSCC	Retrained dSCC
GM12878 (train)	0.892 (in-domain)	
IMR90	0.630	0.795
K562	0.653	0.754

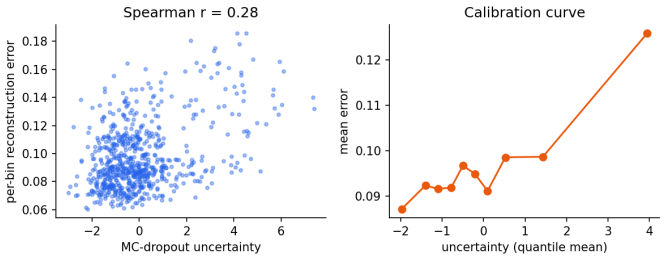


Fig. 3. GM12878 chr22. Left: per-bin uncertainty against error. Right: mean error by uncertainty decile.

The attributions describe what the network uses; they are not evidence that any modality is needed.

V. DISCUSSION AND LIMITATIONS

On dense Hi-C, GenoFold is one of several methods that land within noise of one another, and it is the slowest of them. Its case rests on shallow data. There it keeps close to measured structure while per-matrix optimizers lose it, but the evidence available now credits the learned prior more than the added modalities, and a fair test against a trained Hi-C-only model is still outstanding. The uncertainty estimate separates only the least reliable bins, and the attribution results show sensitivity rather than necessity.

Several limits remain. The objective trains on Hi-C-derived distances, and the independent imaging check covers one 2 Mb region of one chromosome. The comparison of all methods uses GM12878 chr22 only. All cell lines are human, so nothing here tests the species embedding, which is inactive. Bins kept

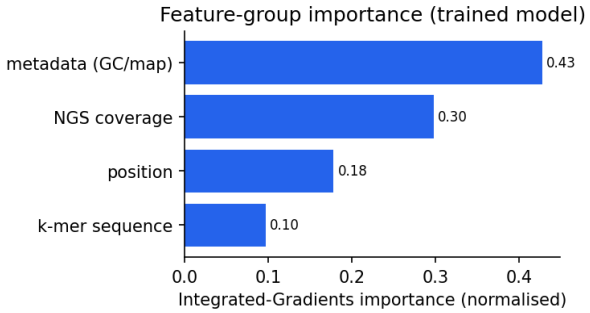


Fig. 4. Integrated-Gradients importance by feature group, trained GM12878 chr22 model.

by the 0.97 filter include low-mappability regions where any reconstruction is weakly constrained.

VI. CONCLUSION

GenoFold reconstructs chromatin structure from Hi-C and depth-independent node features with accuracy level with 3DMax and HiC-GNN on dense data and close to measured distances (Spearman 0.986 and 0.981). With 5% of contacts it stays at 0.978 and 0.969 where 3DMax falls to about 0.82, an advantage we currently attribute to training rather than to multi-modality. The next experiments are a Hi-C-only GenoFold and HiC-GNN on the sparse imaging task, a per-chromosome benchmark, and coverage tests for the uncertainty estimate. Code and checkpoints will be released on acceptance.

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